

STEP III - 1987

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June 18, 2026

Abstract

A collection of solutions to STEP questions for my own use and also for former colleagues and friends. These are very much incomplete. I aim to complete some sporadically although I make absolutely zero promise of doing so with any sort of regular schedule. I largely do this for fun. Although I try my best, I have undoubtedly made several errors so please do use with caution. Any errors, please do let me know at ccbarnesni@gmail.com.

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6. Rearranging the first equation for y and substituting into the 2nd equation gives us:

$$\frac{1}{5} \left(\frac{dx}{dt} + 2x \right) = y$$

$$\frac{dy}{dt} + x - 2y = 2 \cos t$$

$$\frac{1}{5}(\ddot{x} + 2\dot{x}) + x - 2 \left(\frac{1}{5}(\dot{x} + 2x) \right) = 2 \cos t$$

$$\frac{1}{5}\ddot{x} + \frac{2}{5}\dot{x} + x - \frac{2}{5}\dot{x} - \frac{4}{5}x = 2 \cos t$$

$$\ddot{x} + 2\dot{x} + 5x - 2\dot{x} - 4x = 10 \cos t$$

$$\ddot{x} + x = 10 \cos t$$

The complementary function satisfies

$$\ddot{x}_{CF} = -x_{CF} \implies x_{CF} = A \cos(t) + B \sin(t)$$

For the particular integral, we anticipate a solution of the form

$$x = Ct \sin t$$

$$\dot{x} = C \sin t + Ct \cos t$$

$$\ddot{x} = 2C \cos t - Ct \sin t$$

We thus choose $C = 5$. The general solution is then

$$x(t) = A \cos t + B \sin t + 5t \sin t$$

Applying the initial condition $x(0) = 0$ gives $A = 0$. Now substituting this into the equation for y gives us

$$y(t) = \frac{1}{5} \left(B \cos t + (5 + 2B) \sin t + 5t \cos t + 10t \sin t \right)$$

Differentiating yields

$$\dot{y}(t) = \frac{1}{5} \left((10 - B) \sin t + (10 + 2B) \cos t - 5t \sin t + 10t \cos t \right)$$

Applying the initial condition $\dot{y}(0) = 0$ we obtain $10 + 2B = 0$. Thus $B = -5$. The solution is thus

$$\begin{aligned}x(t) &= 5t \sin t - 5 \sin t \\y(t) &= 3 \sin t - t \sin t + 2t \cos t\end{aligned}$$

Note that if instead we have $a > 1$ we will obtain the equation

$$\ddot{x} + (5a - 4)x = 10 \cos t$$

Thus the general solution for the complementary function will instead take the form $x(t) = A \cos \omega t + B \sin \omega t$ for some $\omega > 1$. Similarly, the particular integral will instead take the form $x(t) = C \cos t$.

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